

15. Laplace transform and the transfer function

The Fourier and Laplace transforms look rather similar at first glance. Both involve an integral of a function multiplied by an exponential term over time. However, the Laplace transform is *single-sided*: it only requires integration over values of time greater than zero, and this is an important difference.

Fourier transform. The integral definition is:

$$\mathcal{F}\{f(t)\} = F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt. \quad (15.1)$$

We have transformed a function of time $f(t)$ into a function $F(\omega)$ in the frequency domain. Note that the integral is symmetric in time and involves both positive and negative time.

Laplace transform. The integral definition is:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t) e^{-st} dt. \quad (15.2)$$

Here we say that we have transformed a function of time $f(t)$ into a function $F(s)$ in the s -domain, where s is a *complex* variable:

$$s = \sigma + j\omega. \quad (15.3)$$

The integral is *single-sided*: it only involves values of time greater than zero. Note also that we can recover a Fourier transform by making the substitution $s \rightarrow j\omega$; this is a consequence of the fact that the Fourier transform is a particular case of the Laplace transform, which is the more general treatment.

Physical interpretation of the transforms. In general, when we Fourier-transform a real physical observable, the resulting function also represents a physical observable. For example, when we Fourier-transform a pulse shape $f(t)$, a function of time, we find the spectrum function $F(\omega)$, which we can interpret as the amplitude and phase of all the frequencies in the pulse. In contrast, the Laplace transform does not represent any kind of useful physical observable (although s has units of rad/s). We will use it simply as a tool for solving differential equations. There is one exception to this: s can be used to test for stability in a system. We looked at this briefly in the context of feedback amplifiers, and we will cover this more thoroughly again later in the course. Initially, though, we will see how to use the Laplace transform as a highly effective tool for solving differential equations.

15.1. Fourier analysis of systems

The use of the *Fourier transform* in systems analysis requires the decomposition of an input (“forcing”) function

$y(t)$ into its spectrum function $Y(\omega)$.²⁷ This requires an integral over all time. This means that, in practice, we need to know the value of the forcing function over the entire past history of the system and for all future time. If we do know this, then, given that we know the transfer function of the system $G(\omega)$, we can obtain the spectrum of the system output $X(\omega)$ from the relation

$$G(\omega) = \frac{X(\omega)}{Y(\omega)}.$$

We can then obtain the system response in the time domain by inverse-transforming:

$$x(t) = \mathcal{F}^{-1}\{X(\omega)\}.$$

The *transfer function* $G(\omega)$ contains all the information we need to determine the output. Since it is, in general, a complex function it is more useful to split it into an amplitude $|G(\omega)|$ (called simply the *frequency spectrum*) and a phase term. Although in these lecture notes we focus more on the equivalent transfer function for Laplace transform (as these allow us to solve transient problems), in lab you will work with the Fourier frequency spectrum which is what we tend to measure for periodic signals.

The Fourier approach works perfectly well for systems where the input signal is a continuous repetitive input such as a sinusoid, which for all practical purposes we can take to be infinite in time. This is not just theoretical: in the lab sessions we will see how we can model the transfer functions of the electret microphone, the piezo-sounder, the filter and an amplifier in terms of their frequency response $G(\omega)$. This works because for practical applications we can assume that the input signal is infinite. We know that it is not, but it has existed for long enough that any transient or “start-up” response of the system has long since ceased to be apparent in the output of the system. That is to say, the system output is the same as it would have been had the input existed for all time.

15.2. Advantages of the Laplace transform

When we analyse a complex device in instrumentation, we break the device down into a set of ‘blocks’, each of which can be understood as a linear system and which has its input/output relationship described by a differential equation of the form described in Section 14. When our device is composed of lots of such blocks connected together, and subject to an arbitrary input, we will find that the overall input/output relationship requires us to solve a series of linked differential equations of some considerable complexity. *The usefulness of the Laplace transform lies in its ability to change a differential equation (DE) in the*

²⁷In these notes and in some textbooks the input to a system is labelled $y(t)$ and the output is $x(t)$, which is a little confusing; in other sources this is reversed, so you must be careful... the transfer function is the output divided by the input, essentially a ‘gain’.

time domain into an algebraic expression in the s -domain. This method is generally useful when we wish to find the time-domain behaviour of a system subject to an arbitrary (usually transient) input signal. The general method is as follows:

1. Understand the system behaviour in terms of its governing differential equation, its input (forcing function) and any initial conditions;
2. Transform the DE into the s -domain;
3. Re-arrange algebraically in order to separate the desired output variable;
4. Transform back into the time-domain to discover the resulting behaviour.

For example, we can use the Laplace transform to examine how a system behaves after the application of an impulse input $\delta(t)$ or a step $u(t)$. Recall the thermometer and spring balance examples of Section 14: the Laplace method allows us to determine how the instrument will respond to the input forcing-function. We could of course use general methods for solving ODEs which you have come across before; however, the Laplace method is significantly more powerful.

15.3. Transient analysis

In systems analysis we often want to understand the response of the system to a transient applied at the input at a given time (usually taken to be $t = 0$ for simplicity). This is known as *transient analysis* and is highly applicable for instrumentation applications, since generally we know the state of our instrument now (i.e. the initial conditions), we know the governing differential equations, and we know what the input is. We do not generally know (or care) about the past history of the system. The Laplace transform is single-sided, so only requires integration over positive (future) time, and is generally rather effective at solving these kind of transient problems. For example, we may have a circuit which is either *initially relaxed* (has no charge stored or currents flowing) or it may have *initial conditions* (due to charge stored on capacitors or current flowing in inductors). If at time $t = 0$ we throw a switch to apply some new input signal (forcing function) or even to change the configuration of the circuit by adding capacitance or inductance, then the Laplace transform allows us to solve these sorts of problems. We can evaluate how the system will evolve for all time $t > 0$. This is a physically realistic situation: we know the state of the system now and we apply an input now; we want to observe how the system will evolve. Note that many of these sort of problems can be solved with the Fourier transform too; however, the Laplace method is generally easier and more flexible.

15.4. Properties

The most important properties of the Laplace transform are its linearity and behaviour under differentiation and integration. These properties are described below.

Linearity. Like the Fourier transform, the Laplace transform is a linear operation:

$$\mathcal{L}\{k_1 f_1(t) + k_2 f_2(t)\} = k_1 F_1(s) + k_2 F_2(s). \quad (15.4)$$

First time derivative.

$$\mathcal{L}\left\{\frac{d}{dt}f(t)\right\} = sF(s) - f(0^+), \quad (15.5)$$

where $f(0^+)$ is the initial condition of the system: it is the value of $f(t)$ as $t \rightarrow 0$ from positive t . Recall that we do not need to know (or care) about f for $t < 0$.

Second time derivative.

$$\mathcal{L}\left\{\frac{d^2}{dt^2}f(t)\right\} = s^2 F(s) - sf(0^+) - f^{(1)}(0^+), \quad (15.6)$$

where $f^{(1)}(0^+)$ is the first derivative of $f(t)$ evaluated as $t \rightarrow 0^+$.

Integral with zero initial conditions.

$$\mathcal{L}\left\{\int_0^t f(\xi)d\xi\right\} = \frac{F(s)}{s}. \quad (15.7)$$

Time shifting.

$$\mathcal{L}\{f(t - \lambda)u(t - \lambda)\} = e^{-s\lambda} F(s). \quad (15.8)$$

Frequency shifting.

$$\mathcal{L}\{e^{at}f(t)\} = F(s - a). \quad (15.9)$$

15.5. Elementary Laplace transform pairs

The forward transform is relatively straightforward to derive for many common functions. The inverse transform, by contrast, is given by

$$\mathcal{L}^{-1}\{F(s)\} = f(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s) e^{st} ds, \quad (15.10)$$

and therefore requires integration in the complex plane. Don't try this at home! In order to perform most inverse transforms (and many forward transforms too), it is usually sufficient to use tables of elementary transform pairs²⁸ such as that given in Figure 15.2 at the end of this section.

15.6. Transfer function

Also with Laplace transforms, the *transfer function* is by definition the ratio of the output of a system to its input, where now both are s -domain representations – see Figure 15.1. So, if we have a system where the input is represented by some function $y(t)$ and the output is $x(t)$, then the transfer function is

$$G(s) = \frac{X(s)}{Y(s)} = \frac{\mathcal{L}\{x(t)\}}{\mathcal{L}\{y(t)\}}. \quad (15.11)$$

There are some conditions attached to this definition:

²⁸As a general guide, you should know by heart some of the basic transforms such as entries 1, 2 and 4, while any others you might need would be given on an exam paper.

1. The input and output signals $y(t)$ and $x(t)$ must be continuous functions of time;
2. The system in question must be linear time-invariant;
3. The system must be *initially relaxed* or in a quiescent state. Generally, for the kind of electrical systems we are looking at, this means the system shall not be energised (no energy stored on capacitors or in inductors) at time $t = 0$.

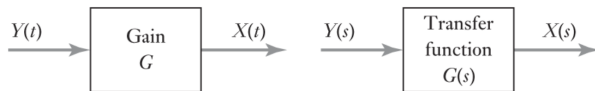


Figure 15.1: The transfer function.

For many ‘complex systems’ – such as an instrument – we can break the design down into component building-blocks such as:

- Sensors;
- Filters;
- Amplifiers (including feedback loops);
- Output transducers.

Typically, each block can be represented as a zero-, first- or second-order LTI system, for which we can individually determine the transfer function according to the definition of equation (15.11). The real power of the transfer function comes when we build, according to simple rules, the transfer function for the whole system. This allows us to find the output of the whole system $x(t)$ in response to any arbitrary input $y(t)$. The rules for this *block diagram algebra* are simple. If we know the transfer functions G_1 and G_2 of two functional blocks, then:

- The transfer function of the *series* (or *cascade*) connection of G_1 followed by G_2 is $G = G_2 G_1$, i.e. the product of the individual transfer functions (note: the ordering can be important!);
- The transfer function for the *parallel* connection is $G = G_1 + G_2$;
- The transfer function of the *feedback* connection, with (open-loop) amplifier block transform G_1 and (negative) feedback block transform $-G_2$ is equal to $G = G_1 / (1 + G_1 G_2)$.

We will see how to apply this method in Section 16.

Entry No.	$f(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st} ds$	$F(s) = \int_0^\infty f(t)\bar{e}^{-st} dt$
1	$\delta(t)$	1
2	$u(t)$	$\frac{1}{s}$
3	$t^n \text{ for } n > 0$	$\frac{n!}{s^{n+1}}$
4	e^{-at}	$\frac{1}{s+a}$
5	te^{-at}	$\frac{1}{(s+a)^2}$
6	$\frac{t^{n-1}e^{-at}}{(n-1)!}$	$\frac{1}{(s+a)^n}$
7	$\frac{1}{b-a} (e^{-at} - e^{-bt}) a \neq b$	$\frac{1}{(s+a)(s+b)}$
8	$-\frac{1}{b-a} (ae^{-at} - be^{-bt}) a \neq b$	$\frac{s}{(s+a)(s+b)}$
9	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
10	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
11	$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2 + \omega^2}$
12	$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2 + \omega^2}$
13	$\sinh \omega t$	$\frac{\omega}{s^2 - \omega^2}$
14	$\cosh \omega t$	$\frac{s}{s^2 - \omega^2}$
15	$\frac{\sqrt{a^2 + \omega^2}}{\omega} \sin(\omega t + \phi), \phi = \tan^{-1} \frac{\omega}{a}$	$\frac{s+a}{s^2 + \omega^2}$
16	$\frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1-\zeta^2} t) \quad \zeta < 1$	$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$
17	$\frac{1}{a^2 + \omega^2} + \frac{1}{\omega\sqrt{a^2 + \omega^2}} e^{-at} \sin(\omega t - \phi),$ $\phi = \tan^{-1}\left(\frac{\omega}{-a}\right)$	$\frac{1}{s[(s+a)^2 + \omega^2]}$
18	$1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n \sqrt{1-\zeta^2} t + \phi),$ $\phi = \cos^{-1} \zeta, \zeta < 1$	$\frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$

Figure 15.2: Elementary Laplace transform pairs (from Poularikas and Seeley).

16. Solving problems with the transfer function

Laplace transforms allow us to find the time response of complex systems driven by an arbitrary forcing function, often without having to directly formulate and solve the differential equation. The method works well for problems which can be formulated to start at some given time, usually taken to be $t = 0$ for simplicity. This is what we actually deal with on most experiments. We set up some apparatus and then throw a switch at a starting time $t = 0$. We then want to know what happens for positive time. To formulate and solve Laplace transforms for simple input functions we can look up the relevant time domain functions and their s -domain transforms in a table (see Figure 15.2). For a more complex system we will probably need to simplify the Laplace transform representation somewhat before we can transform back into the time-domain. The technique we quite commonly use to do this involves partial fractions (see last page in this section).

Note that the Laplace transform of the delta function $\delta(t)$ is 1. This means that the time response of a system subject to an impulse at its input is given by the inverse Laplace transform of the transfer function, i.e. $x(t) = \mathcal{L}^{-1}\{G(s)\}$ for a δ -function input. That is to say, *the impulse response – which we may call $g(t)$ – is the time-domain representation of the transfer function.* This is one of the reasons why the delta function is so commonly used in analysis of systems. Another way of understanding this is to recall that the Fourier transform of a δ -function is an infinite flat spectrum, so a perfect impulse contains all frequencies in equal amounts. In other words, when we apply an impulse to a system, we stimulate it with all frequencies *simultaneously* and hence we can measure its frequency response – rather than some property of the input pulse. In practical applications, such as the Elvis Bode Analyser, it is inefficient to stimulate the system with all frequencies at once (and indeed impossible to generate a mathematically *true* impulse, so the Bode Analyser sweeps the frequency input in order to buildup the frequency response.

16.1. Step-input response for a series system

Take an instrumentation system constructed from several stages as shown in Figure 16.1; we wish to determine its response to a unit-step applied to the input.

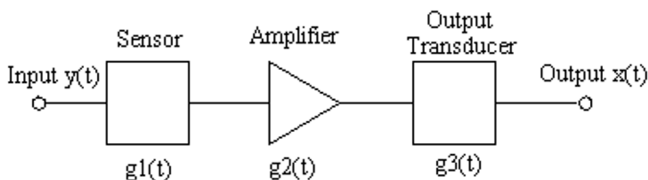


Figure 16.1: System consisting of three functional blocks in series.

We begin by determining the individual responses of the sensor, amplifier and transducer to the $\delta(t)$ function.

This is the all-important *impulse response function* which can be determined either analytically or experimentally.

For the instrument in the figure let us say we determine the impulse responses in the time domain, that is to say we have measured the input and output responses of each transducer to impulse inputs. As mentioned previously, this is not always the most efficient method, but it is sometimes used in reality, especially for certain mechanical/acoustic transducers or for photomultiplier tubes (the single photoelectron response is essentially the impulse response function of a PMT). In this case let us say we measure the impulse responses to be:

- Sensor: $g_1(t) = e^{-t}$;
- Output transducer: $g_3(t) = e^{-2t}$.

Let us further say that we have tested the amplifier and found it to have a voltage gain of 4. If we assume that it is an ideal amplifier then (as a zero-order system) it will have an impulse response given by:

- Amplifier: $g_2(t) = 4\delta(t)$.

What is the output of the system as a function of time for, say, an input step $y(t) = u(t)$? First we need to formulate a transfer function for the complete system. We know the time domain responses of the individual components and from these we can find their transfer functions. From our table of Laplace transforms in Figure 15.2,

$$\begin{aligned} G_1(s) &= \frac{1}{s+1} ; \\ G_2(s) &= 4 ; \\ G_3(s) &= \frac{1}{s+2} . \end{aligned}$$

Now we can find $G(s)$, the transfer function for the whole system:²⁹

$$\begin{aligned} G(s) &= G_1(s)G_2(s)G_3(s) \\ &= \frac{4}{(s+1)(s+2)} . \end{aligned}$$

The input to the system is a unit step function $y(t) = u(t)$, and the Laplace transform of this is simply $1/s$. To find the time response of the system we recall that $X(s) = G(s)Y(s)$, where $X(s)$ and $Y(s)$ are the Laplace transforms of the output and input of the system and $G(s)$ is the transfer function; so,

$$\begin{aligned} x(t) &= \mathcal{L}^{-1}\{X(s)\} \\ &= \mathcal{L}^{-1}\{G(s)Y(s)\} \\ &= \mathcal{L}^{-1}\left\{\frac{4}{s(s+1)(s+2)}\right\} . \end{aligned}$$

²⁹Note that the order for this cascade system should be more correctly $G = G_3G_2G_1$, but for most LTI systems with scalar transform functions the terms clearly commute. We need to be careful, for example, if one block *loads* the next...

Our task now is to evaluate the inverse Laplace transform of this function, and we do this for our example using partial fractions. We write our s -domain function as a sum of partial fractions (see the final page in this section):

$$X(s) = \frac{4}{s(s+1)(s+2)} = \frac{A_0}{s} + \frac{A_1}{s+1} + \frac{A_2}{s+2}.$$

This is an identity and valid for all values of s , therefore we can choose values of s which make some of the constants disappear. Multiplying through by s and evaluating for $s = 0$ gives

$$\left| \frac{4}{(s+1)(s+2)} = A_0 + \frac{A_1 s}{s+1} + \frac{A_2 s}{s+2} \right|_{s=0},$$

$$A_0 = 2.$$

Similarly, for A_1 we can multiply through by $s+1$ and evaluate for $s = -1$:

$$\left| \frac{4}{s(s+2)} = \frac{A_0(s+1)}{s} + A_1 + \frac{A_2(s+1)}{s+2} \right|_{s=-1},$$

$$A_1 = -4.$$

And for A_2 we can multiply through by $s+2$ and evaluate for $s = -2$:

$$A_2 = 2.$$

Putting these constants back into our identity for $X(s)$ we obtain

$$X(s) = \frac{2}{s} - \frac{4}{s+1} + \frac{2}{s+2}.$$

This is now in a form that we can simply write down a time response for, with reference to our Laplace transform tables. If we do this for the individual components of $X(s)$ we find, for positive times,

$$x(t) = 2 - 4e^{-t} + 2e^{-2t}.$$

This function is plotted in Figure 16.2 for $0 < t < 8$ seconds. It shows (nearly) the sort of time behaviour we expect from first order systems subject to a step input (or a heavily-damped second order system). The response to the step is an initial rise, which then slows down, and tends to some steady state value for large t . Note that transfer functions of the form $G_1(s)$ and $G_3(s)$ are characteristic of first-order systems while the ideal amplifier (zero-order) has no time-dependence and hence its transfer function is a constant.

16.2. Impulse response with known transfer function

We now attempt to find the time-domain impulse response of a system with a (known) transfer function

$$G(s) = \frac{s+3}{(s+2)^2}.$$

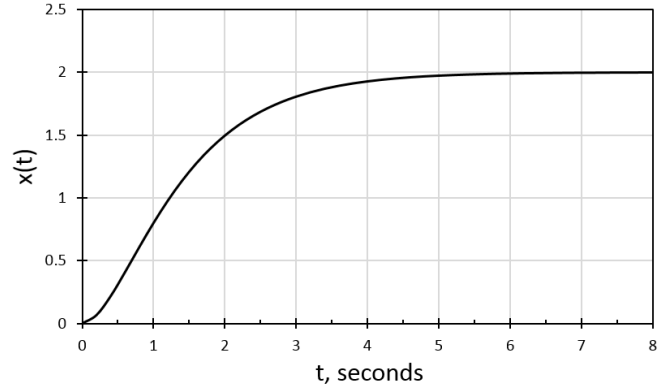


Figure 16.2: System response to a step-input $y(t) = u(t)$.

To find the time-domain response of the system we note that $g(t) = \mathcal{L}^{-1}\{G(s)\} = \mathcal{L}^{-1}\{X(s)\}$ (since $y(t) = \delta(t)$ and, hence, $Y(s) = 1$) and so

$$g(t) = \mathcal{L}^{-1}\left\{\frac{s+3}{(s+2)^2}\right\}.$$

As before, we need a partial fraction expression to simplify the inverse transform. For denominators of order > 1 we must write the partial fraction as

$$\frac{s+3}{(s+2)^2} = \frac{A_2}{(s+2)^2} + \frac{A_1}{s+2}.$$

To find A_2 we multiply $G(s)$ by $(s+2)^2$ and set $s = -2$ to obtain $A_2 = 1$. We now need to find A_1 ; however, there is a problem with continuing the method. To find this coefficient we cannot use $s = -2$ again as we have already used this to find A_2 . Instead, we can *differentiate* $G(s)(s+2)^2$:

$$\frac{d}{ds}G(s)(s+2)^2 = \frac{d}{ds}(s+3) = \frac{d}{ds}(A_2 + A_1(s+2)).$$

$$1 = A_1.$$

Therefore,

$$G(s) = \frac{1}{(s+2)^2} + \frac{1}{s+2},$$

which can be easily inverse-transformed into

$$g(t) = t e^{-2t} + e^{-2t}.$$

16.3. Other applications

We have seen here how Laplace methods allow complex systems to be solved by transforming differential equations in the time-domain into algebraic expressions in the s -domain. The fact that the Laplace transform is single-sided and takes into account initial conditions means that it is very well suited for solving transient analysis problems in mechanical and electrical engineering. These methods

are applicable across the physical and engineering disciplines, and biological systems (including complex feedback loops) can be also modelled. Other applications are in economics, where the parameter to be determined could be price and the constants relate to supply and demand. In general, any continuous-time system represented by ordinary differential equations can be studied, especially where we want to set the system up in some starting configuration and observe how it evolves with time.

As a final note, you should be aware that there is an equivalent to the Laplace transform for discrete (quantised) signals called the Z -transform. This is of central importance in the field of digital signal processing, though we will not be able to cover this in the course.

16.4. Partial fractions

Consider the identity

$$\frac{2}{5} - \frac{3}{4} + \frac{1}{2} = \frac{8 - 15 + 10}{20},$$

where 20 is the lowest common multiple. *Partial fraction decomposition* is the *reverse* of this process. In general, any rational function

$$\frac{P(s)}{Q(s)}$$

can be re-written using partial fractions. First, it is essential to check that the degree of the numerator is less than the degree of the denominator. If not, it is necessary to first divide-out by long division. This is tedious so we will avoid it in this course, and in general the sort of real-world problems we will encounter naturally lead a higher degree of s in the denominator. We proceed as follows.

For each factor of $Q(s)$ in the form $(as + b)^m$ introduce terms

$$\frac{A_1}{as + b} + \frac{A_2}{(as + b)^2} + \dots + \frac{A_m}{(as + b)^m}.$$

For factors of the form $(cs^2 + ds + e)^n$ introduce terms

$$\frac{C_1s + D_1}{cs^2 + ds + e} + \frac{C_2s + D_2}{(cs^2 + ds + e)^2} + \dots + \frac{C_ns + D_n}{(cs^2 + ds + e)^n}.$$

There are various methods to solve for the constants, but remember that

$$\frac{P(s)}{Q(s)} = \frac{A_1}{as + b} + \frac{C_1s + D_1}{cs^2 + ds + e} + \dots$$

is an identity valid for all values of s . We can rearrange and substitute values of s which cause some unknowns to disappear. For example, in

$$X(s) = \frac{s + 3}{(s + 2)^2} = \frac{A_1}{s + 2} + \frac{A_2}{(s + 2)^2},$$

we can multiply $X(s)$ by $(s + 2)^2$ and set $s = -2$ to get A_2 . To get A_1 we can differentiate $X(s)(s + 2)^2$ with respect to

s and then use $s = -2$ a second time. Alternatively, we can gather terms in powers of s to obtain a set of simultaneous equations

$$X(s)(s + 2)^2 = s + 3 = A_1(s + 2) + A_2.$$

Gather terms:

$$0 = s(A_1 - 1) + (A_2 + 2A_1 - 3),$$

from which

$$\begin{aligned} A_1 - 1 &= 0 \\ A_2 + 2A_1 - 3 &= 0. \end{aligned}$$

Whichever method is best depends on the form and complexity of the functions.